

ZAGREB MAKSIMIR, Croatia

WMO: 142400

Lat: 45.8219N

Lon: 16.0336E

Elev: 420

StdP: 14.47

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.3	19.9	8.6	8.7	17.2	13.4	11.0	24.2	16.0	30.9	13.7	32.4	3.6	30	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	91.1	71.1	88.2	70.1	85.3	68.9	72.9	86.4	71.5	84.8	70.1	82.7	4.9	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
68.5	106.4	79.0	66.9	100.6	77.6	65.6	95.8	76.2	36.7	86.3	35.4	84.9	34.2	83.0	80.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.5	10.5	9.1	9.3	95.0	6.2	3.9	4.9	97.8	1.2	100.1	-2.3	102.2	-6.8	105.0		
(5)			8.7	75.8	6.0	1.9	4.4	77.2	0.8	78.3	-2.5	79.3	-6.9	80.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	53.8	34.4	38.0	45.5	54.2	62.3	69.0	72.0	71.5	62.6	54.1	44.9	35.8
	DBStd	15.05	8.20	8.23	7.81	6.78	6.21	6.20	5.46	5.51	6.05	7.22	7.94	7.83
	HDD50	1724	487	341	180	38	2	0	0	0	43	189	444	
	HDD65	4783	950	756	604	328	128	31	8	10	116	343	603	906
	CDD50	3099	2	5	41	163	382	570	681	668	379	169	36	3
	CDD65	682	0	0	0	2	43	151	224	213	45	4	0	0
	CDH74	6537	0	0	1	51	427	1466	2207	1992	361	32	0	0
	CDH80	2261	0	0	0	5	79	482	846	776	73	0	0	0
(14) Wind	WSAvg	3.6	3.4	3.8	4.5	4.2	4.1	3.7	3.7	3.4	3.3	3.0	3.2	3.1
(15) Precipitation	PrecAvg	39.4	4.1	2.2	3.5	2.6	3.1	3.4	2.9	3.6	3.5	3.7	3.8	3.9
	PrecMax	69.6	28.0	6.9	15.7	5.6	8.6	7.2	7.3	10.4	8.4	12.0	9.3	15.9
	PrecMin	22.0	0.1	0.2	0.4	0.6	0.7	0.7	1.0	0.6	1.2	0.2	0.4	0.4
	PrecStd	12.8	5.7	1.6	3.3	1.2	2.0	1.5	1.3	2.0	1.7	2.5	2.3	3.9
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.1	63.6	71.4	79.7	85.8	92.1	95.2	96.8	86.6	77.6	68.8	59.8
		MCWB	50.2	50.2	54.6	60.7	66.7	71.6	71.5	70.1	67.7	64.1	57.6	52.2
	2%	DB	54.7	58.5	66.8	74.5	82.1	88.6	91.1	91.1	81.8	73.5	63.6	54.9
		MCWB	48.0	48.4	52.7	57.9	65.0	70.6	71.6	70.6	66.9	62.5	55.5	49.3
	5%	DB	51.0	54.8	62.7	71.1	78.9	85.6	87.8	87.9	78.2	69.5	59.8	51.2
		MCWB	45.9	46.7	50.9	56.4	63.4	69.3	70.4	69.7	65.2	60.4	53.8	46.9
	10%	DB	46.8	51.1	58.8	67.5	75.4	82.4	84.8	84.2	74.6	65.5	56.5	47.0
		MCWB	43.0	44.5	49.2	54.7	61.8	67.7	69.0	68.8	63.3	58.7	52.4	44.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.7	55.5	62.2	69.0	74.4	75.3	74.6	70.0	65.5	58.9	52.9
		MCDB	57.2	61.3	68.3	76.9	82.4	87.6	89.3	88.7	82.1	75.0	65.1	58.2
	2%	WB	48.6	49.3	53.5	59.4	66.6	72.2	72.9	72.5	67.8	63.3	56.6	50.0
		MCDB	53.9	57.3	64.6	71.6	79.1	85.3	86.7	86.2	79.0	71.3	62.0	54.6
	5%	WB	46.0	47.3	51.8	57.4	64.6	70.4	71.4	70.9	66.2	61.3	54.6	47.2
		MCDB	50.4	53.8	60.5	67.6	76.3	82.9	85.0	84.3	75.9	67.8	59.4	50.7
	10%	WB	43.2	45.0	50.0	55.7	62.7	68.7	70.0	69.5	64.5	59.4	52.3	44.3
		MCDB	46.5	50.3	58.1	65.4	73.5	80.6	82.6	82.0	72.6	64.8	56.4	46.9
(35) Mean Daily Temperature Range	5% DB	MDBR	11.4	14.6	17.3	19.2	19.2	19.4	20.0	20.2	17.7	16.2	12.5	10.6
		MCDBR	18.8	22.6	24.8	26.5	24.8	24.0	25.1	26.4	23.5	21.8	19.2	17.6
		MCWBR	12.4	13.6	13.1	12.4	10.6	9.6	9.2	9.5	10.3	11.7	11.7	12.3
	5% WB	MCDBR	17.2	20.3	21.9	22.9	22.7	22.3	22.9	23.4	21.1	19.8	17.3	16.4
		MCWBR	11.6	12.7	11.9	11.0	10.2	9.5	9.2	9.6	9.8	11.0	11.0	11.8
(40) Clear-Sky Solar Irradiance	taub		0.335	0.361	0.398	0.447	0.435	0.439	0.439	0.440	0.425	0.397	0.363	0.335
	taud		2.348	2.287	2.229	2.112	2.195	2.228	2.227	2.237	2.246	2.305	2.350	2.385
	Ebn at Noon		238	254	261	257	265	264	262	257	249	238	225	225
	Edh at Noon		24	31	38	47	45	43	43	41	37	31	24	22
(44) All-Sky Solar Radiation	RadAvg		390	632	1010	1385	1714	1926	1936	1703	1178	770	426	318
	RadStd		59	106	138	158	158	139	127	185	152	89	58	49

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+1.15	N/A	N/A	+2.39	+1.12	N/A	N/A	N/A	N/A	N/A
Regional (54 neighbors)	+0.81	N/A	N/A	+1.44	+0.81	+0.68	N/A	-209	+155	+101

Nomenclature: See separate page